物联网信息处理技术

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1.1

Data Attributes:

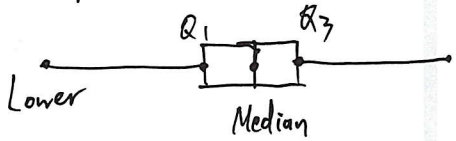
Types:

- Nominal 命名型 {black, blond, brown}
- Binary (0 & 1)
  - ① Symmetric binary 对称 (gender)
  - ② Asymmetric binary 不对称 (positive & negative)
- Numeric 数值型: Interval scaled, Ratio-scaled
- Ordinal 顺序型: {small, medium, large}

计算 median: $median = L_1 + \left(\frac{\frac{n}{2} - (\sum freq)_l}{freq_{median}}\right) \cdot width$

L_1 上一个范围右端点, $\frac{n}{2}$: 所有数字/2, $(\sum freq)_l$ 在此之前范围和, width 当前取值范围

Dispersion of Data (Blox plot)



$IRQ = Q_3 - Q_1$

Q_1: 25% 分位数
Q_3: 75% 分位数.

Measuring Data Similarity

- Nominal attributes: $d(i,j) = \frac{p-m}{p}$ p: 属性个数 m: i和j匹配上的属性个数.
- Binary attributes:

	j	
	1	0
i 1	q	r
0	s	t

对称: $d(i,j) = \frac{r+s}{q+r+s+t}$

非对称 $d(i,j) = \frac{r+s}{q+r+s}$

$sim(i,j) = \frac{q}{q+r+s}$

Distance for ordinal attributes

$r_{if} \in \{0, 1, 2, \cdots, M_f\}$ 按顺序 $z_{if} = \frac{r_{if} - 1}{M_f - 1}$

☆ Data preprocessing 4个步骤:

① Data cleaning: fill in missing value, smooth data, identify outliers
② Data integration: Integrate data from multiple databases
③ Data reduction: Dimentionality reduction 降维、Numerosity reduction
④ Data transformation: Normalization, Concept hierachy generation

Handle Noisy data?

- Binning 装箱
  equal-width $W = \frac{Max-Min}{N}$ 以等宽划分数据
  equal-depth：每个类数据个数相同（先排序）
  bin boundary 每个类里的数换成边界（离哪个近用哪个）
  bin means：—— - ---- — 均值

- Curse of dimensionality 维度诅咒：
  When dimension increases, data become sparse. density and distance between points, which useful becomes less meaningful. combinations increase.
  方法：维度降低、attribute creation、Numerosity Reduction
- Sampling：obtaining a small samples to represent whole dataset N.

Data Transformation：

- Normalization：

① Min-Max 归一化
把一个范围在 [a,b] 转化为 [c,d]，原先 $x \in [a,b]$ 变成？

$$x' = \frac{x-a}{b-a}(d-c) + c$$

② z-score 归一化：$v' = \frac{v-\mu}{\sigma}$

③ 十进制归一化 $v' = \frac{v}{10^j}$ $j = \min_j \left(\max(|v|) < 1\right)$

Chapter 2 Data Mining

2.1

Frequent pattern：a pattern that occurs frequently in a dataset.

k-item set：$X = \{x_1, x_2, \dots, x_k\}$

Association rules：

support $(X \to Y) = P(X \cup Y)$ confidence $(X \to Y) = P(Y|X)$

☆ Closed Patterns and Max Patterns ↗无超集支持度相同

closed pattern：if X is frequent and no $Y \supset X$ with same support as X

max pattern：An itemset is max if X is frequent and no frequent super pattern $Y \supset X$ exist. 不存在更大的频繁项集

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可扩展的频繁项集挖掘

Apriori 每轮计算单项的频率

① self-joining L_k → $min_sup = k$

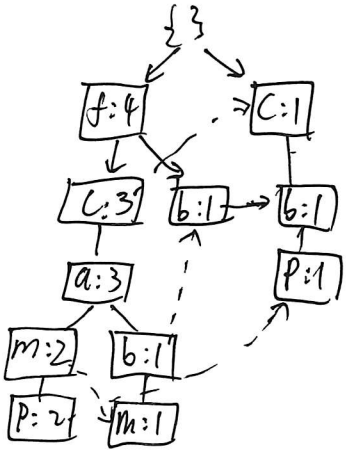
② pruning 剪枝 → $C_3 = C_2 \bowtie C_2$

① get 1-itemset ② generate k+1 from k ③ Test

FP Graath 深搜

① scan DB find frequent 1-itemset
② sort items in decending order
③ Scan again, construct FP-tree

} 构建树



F-list: f-c-a-b-m-p

划分子集: 有p / 有m无p ··· 有c无abmp / f

transfer prefix path to form conditional pattern base:

	Cond. pattern base
c	f:3
a	fc:3
b	fca:1 f:1 c:1
m	fca:2, fcab:1
p	fcam:2, cb:1

from pattern base to conditional FP-tree:

① accumulate count for each item in base
② FP-tree

→

m: conditional FP-tree

{}
|
f:3
|
c:3
|
a:3

m

fm cm am
fcm fam cam
fcam

Item	Conditional pattern base	Conditional FP-tree
p	(fcam:2)(cb:1)	{(c:3)}|p
m	(fca:2) (fcab:1)	{f:3, c:3, a:3}|m
b	(fca:1) (f:1) (c:1)	∅
a	(fc:3)	~~fc:3~~ {f:3, c:3}|a
c	(f:3)	{(f:3)}|c
f	∅	∅

优势: completeness
compactness

伪代码：(FP-Growth)

Input：D (database)，min_sup　Output：patterns

(a) scan D once, collect F, 1-itemset and their counts

(b) create root for Tree, label as null. For each transaction Trans in D,
select and sort frequent items in Trans according of L
Let the sorted item list be [p|P]　P：remaining list　p：first item
Call insert_tree([p|P],T)

ECLAT：

Mining can be performed on dataset by intersecting TID sets of every pairs

$lift = \frac{P(AB)}{P(A)P(B)}$　才是升度

2.2 Classification

2-step process：

① Model construction (training set，描述-组类)

② Model usage

- Accuracy rate：test set independent from training set

Decision Tree

branch：outcome　leaf node：holds a class label

Induction：自顶向下递归

Entropy：$H(Y) = -\sum_{i=1}^{m} p_i \log_2(p_i)$　$Y=\{y_1, \cdots, y_m\}$

$H(Y|X) = -\Sigma_x P(x) \cdot H(Y|X=x)$

Attribute selection：highest Info gain

$g(D,A) = H(D) - H(D|A)$

Bayes

$P(H|X) = \frac{P(X|H) \cdot P(H)}{P(X)}$　困难：require initial knowledge

Max：$P(C_i|X)$　$P(C_i|X) = P(X|C_i) \cdot P(C_i)$

Zero_Prob：each case +1

Model evaluation：

Actual \ Predict	T	F
T	TP	FN
F	FP	TN

$Precision = \frac{TP}{TP+FP}$

$Recall = \frac{TP}{TP+FN}$

数据集划分：

Holdout：{Train (2/3) / Test (1/3)} Random sampling ↺ 重复k次取平均

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Cross-validation (k folds) k个互斥子集

Boot_strap：小数据集 (.632) $Acc(M)=\frac{1}{k}\sum_{i=1}^{k}(0.632\times Acc(M_i)_{test}+0.368\,Acc(M_i)_{tr})$

2.3 Cluster：a collection of data objects

Quality：high intra similarity，low inter-class

Partitioning：n个object ⇒ k个clusters

$E=\sum_{i=1}^{k}\sum_{p\in C_i}(d(p,c_i))^2$ c_i is the centroid 质心 of C_i

K-Means：

① 划分k个非空集 ② 计算种子点(质心)平均 ③ 将每个点划分到邻近中 ④ 返回2无更改

Input：k, D Out：k个clusters

Method：① 选k个点质心 ⇒ 划分 ② update (mean) until no change $O(tkn)$

问题：sensitive to outliers、预先定义k

K-Medoids of PAM

用中心点(medoid)代替mean作为reference point.

① 选k个点 ② 划分邻近 ③ 随机选一个非中心 O_{random} 计算交换Loss

④ 若Loss<0 则交换 O_j 与 O_{random} 直至不变 $(S=E_{swapping}-E_j)$

Hierachical Methods 分层聚类

AGNES (聚合嵌套)：从小的单元往上聚合

DIANA：上面的逆序(分裂)

Distance between clusters：single link：$d(k_i,k_j)=\min(t_{ip},t_{jq})$ Avg：$dist(k_i,k_j)=avg(t_{ip},t_{jq})$

Medoid：$d=dist(M_i,M_j)$ complete link：$=\max(t_{ip},t_{jq})$ Centroid：$=dist(C_i,C_j)$ 平均值

Density-based Methods

建模 dense region

Eps：邻域最大半径 MinPts：该点Eps邻域内最小点数

core object：邻域内点数≥MinPts

Directly density-reachable：$p\in N_{eps}(q)$，$|N_{eps}(q)|\geq MinPts$

density-reachable(p,q)：$p\to r\to s\to\cdots\to q$ 中间都是directly...

density-connected(p,q)：$\exists o$，$p\leftrightarrow o$，$o\leftrightarrow q$ 是density_reachable

DBSCAN：发现核心对象

Input：D、ε、MinPts

Method： mark all unvisited

① random select one point, mark p is visited.

if ε- neighbour of p ≥MinPts

create new cluster C, add p to C. let N be objects in p's neighbour

for p' in N, if p'!= visited, mark p' visited; if Np' ≥ MinPts, add p' neighbour as to N.

if p' isn't a member, add p' to C. end Output C.

else mark p as noise.

until no object is unvisited.

evaluate cluster： $k \approx \sqrt{\frac{n}{2}}$

silhouette coefficient $s(o) = \frac{b(o)-a(o)}{\max\{a(o),b(o)\}}$

$a(o) = \frac{\sum_{o' \in C_i, o \neq o'} dist(o,o')}{|C_i|-1}$ $\quad b(o) = \min_{C_j : 1 \le j \le k, j \neq i} \left\{ \frac{\sum_{o' \in C_j} dist(o,o')}{|C_j|} \right\}$

$s(o)$ 接近1 表示紧凑

2.4 Outlier

定义：a object deviates significantly from normal objects

type：
- Global outlier：明显偏离其它
- context outlier：上下文、行为属性
- collective outlier：子集

Proximity-Based Approaches

· Assumption of proximity-based：离群值接近程度的异常

① Distance-based outlier detection

No ≤ ε 一定是outlier

o is DB(r, π) outlier if $\frac{\|\{o' | dist(o,o') \le r\}\|}{\|D\|} \le \pi$

Input： $D=\{o_1, \cdots, o_n\}$, $r(>0)$、π $(0<\pi\le 1)$ Output：DB(r, π)

Method：
```
for i=1, ..., n
    cnt=0
    for j=1 to n do
        if i≠j and dist(Oi, Oj) ≤ r then
            cnt+=1;
            if cnt ≥ π·n, then exit.

    print Oi
end.
```

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② Density-based outlier detection
离群点密度

k-distance neighborhood of o,

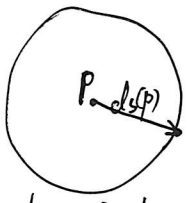
$N_k(o) = \{o' \mid o' \in D, dist(o,o') \le dist_k(o)\}$

$dist_k(o)$：o到第k近点的距离

③ Local Outlier Factor (LOF)

可达距离：$reachdist_k(o \leftarrow o') = \max\{dist_k(o), dist(o,o')\}$

局部可达密度：$lrd_k(o) = \dfrac{\|N_k(o)\|}{\sum_{o' \in N_k(o)} reachdist_k(o' \leftarrow o)}$



$|N_k(p)| \ge k$.

$LOF_k(o) = \dfrac{\sum_{o' \in N_k(o)} \frac{lrd_k(o')}{lrd_k(o)}}{\|N_k(o)\|}$ 局部离群因子

$LOF < 1$：密集点.

Chapter 3 Information Retrieval

3.2 Modeling

IR model ⇒ ranking function

a ranking is an ordering of the documents reflect the relevance

$[D, Q, F, R(q_i, d_j)]$

1. D is documents Q is queries F : framework R : ranking function

index term: word or a group of consecutive words

$V = k_1\ k_2\ k_3 \cdots k_t$ ← vocabulary 词汇表

conjunctive component $111\cdots100\cdots0$

文档 $d_j/q \Rightarrow c(d_j)/c(q)$ unique conjunctive component

term ←	d_1	d_2 ← document
k_1	f_{11}	f_{12}
k_2	f_{21}	f_{22}
k_3	f_{31}	f_{32}

(k_i, d_j) (k_i, q)

$W_{ij} \in \{0,1\}$ $W_{iq} \in \{0,1\}$.

Boolean Model

Queries : $q = k_a \wedge (k_b \vee \neg k_c)$

Vocabulary : $V = \{k_a, k_b, k_c\}$,

$c(d_j) = (1,1,1,0)$ 只有 $k_a\ k_b\ k_c$

$$sim(d_j, q) = \begin{cases} 1 & \exists\, c(q) \mid c(q) = c(d_j) \\ 0 & else \end{cases}$$

缺点：no partial matching, No ranking

3.2.2 Term Weighting $V=\{k_1,\cdots,k_t\}$ $\vec{d_j}=(w_{1,j},\cdots,w_{t,j})$.

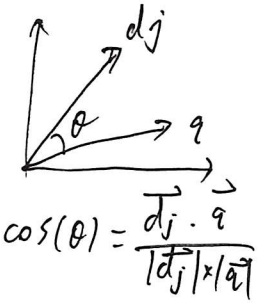
k_i 不在 d_j 中 $w_{ij}=0$ else $w_{ij}\geq 0$

f_{ij} 为 k_i 在 d_j 中出现频率/次数

total frequency of occurance: $F_i=\sum_{j=1}^{N} f_{i,j}$ N为文档量

一般相互独立

document length: $|\vec{d_j}|=\sqrt{\sum_{i}^{t} w_{ij}^2}$



$\cos(\theta)=\frac{\vec{d_j}\cdot\vec{q}}{|\vec{d_j}|\times|\vec{q}|}$

3.2.4 Vector model

$$sim(d_j,q)=\frac{\sum_{i=1}^{t} w_{ij}\times w_{i,q}}{\sqrt{\sum w_{ij}^2}\cdot\sqrt{\sum w_{iq}^2}}$$

☆ tf-idf weights

$$w_{i,q}=(1+\log f_{i,q})\times\log\frac{N}{n_i}$$

$$w_{i,j}=(1+\log f_{i,j})\times\log\frac{N}{n_i}$$

eg: q= to do

① ② $n_1=2$ $n_2=3$ $N=4$ 以 d_1 为例:

① to: $f_{i,q}=1$ $w_{i,q}=(1+\log_2 1)\times\log_2\frac{4}{2}=1$

$j=1$: $f_{i,j}=4$ $w_{ij}=(1+\log_2 4)\times\log_2\frac{4}{2}=3$

② do: $f_{i,q}=1$ $w_{i,q}=(1+\log_2 1)\times\log_2\frac{4}{3}=0.415$

$f_{i,j}=2$ $w_{ij}=(1+\log_2 2)\times\log_2\frac{4}{3}=0.830$

∴ 对 d_1 而言: $sim(d_1,q)=\frac{1\times 3+0.415\times 0.83}{\sqrt{10}\cdot\sqrt{2^2+1^2}}=0.660$.

每个都一样

优点: quality↑, 部分匹配

3.3 Text classification

3.4 Searching

Inverted index: a word-oriented mechanism for indexing a text collection to speed up the searching task.

Signature Files

slower than inverted indexes. "W & Bi = W"

4.1 Image

4.1.2 Image representation (gray-scale)

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sampling points ⇒ grid

1个点：pixel or element

$k = 2^b$ b位，k个区间

$D_E = L_2$ $D_4 = L_1$ $D_8 = L_\infty$

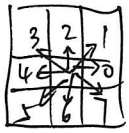
pixel adjacency：4⇒  4邻接 8⇒ 

Contrast（对比度）：物体平均亮度与背景比值

addictive noise：$f(x,y) = g(x,y) + v(x,y)$

Multiplicative：$f = gv$

0：黑 255：白

chain code 

Local preprocessing (filtering)

$\frac{1}{9}\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix}$ 均值 $\frac{1}{8}\begin{bmatrix}1&1&1\\1&1&1\\1&0&1\end{bmatrix}$ 去黑白点

Low-pass filter：softened ~~edges~~ edges.

edge：pixel locations of abrupt luminance change